



15 September 2026

Dear Editors,

We are pleased to submit our manuscript entitled “Cross-border offshore hydrogen trade and carbon emission mitigation in Europe” for consideration as an Article in Nature Communications.

Ireland and the UK have ambitious offshore wind development plans, reaching 37 and 125 GW, respectively, by 2050. Their relatively limited domestic energy demand constrains how much of this offshore wind generation can be absorbed, even with improved system flexibility, while demand centres in continental Europe require increasing supplies of low-carbon fuels. Converting surplus offshore wind into green hydrogen and its derivatives for export could connect these resources with European demand, ease domestic wind-integration pressures and support decarbonisation across borders. However, the roles of Ireland and the UK remain insufficiently quantified when offshore hydrogen is assessed mainly through production costs or individual-country studies alone.

Our study evaluates these roles through an integrated European assessment from 2030 to 2050. We first compare offshore hydrogen production costs and supply potential across major European coastal countries. We then assess domestic offshore electricity and hydrogen integration through physics-constrained hourly power–gas system optimisation, using Ireland as the detailed case study, and link the resulting export-potential estimates to an international hydrogen and ammonia market-clearing model that allocates supply to demand across countries subject to production and transport constraints. This framework brings together production competitiveness, domestic energy needs, shipping costs and external import competition to quantify cross-border flows and their contributions to European decarbonisation.

Our results project the UK to remain the largest net exporter of offshore green hydrogen among the modelled European coastal countries from 2030 to 2050, with Ireland nearly matching it in 2050 at approximately 162 TWh of net exports per year each. Both countries occupy intermediate production-cost positions, showing how substantial offshore resources and limited domestic absorption can support major export roles even without the lowest production costs. The findings identify opportunities for west-to-east renewable-energy trade and clarify how offshore hydrogen exports can support domestic wind integration alongside decarbonisation elsewhere in Europe.

We believe this work is relevant to the broad readership of Nature Communications because it connects offshore renewable-energy development, coupled energy systems and international trade to address a shared decarbonisation challenge. By quantifying how surplus renewable resources in one country can serve energy needs in another, the study informs coordinated investment in offshore generation, hydrogen infrastructure and cross-border supply chains. It will be of interest to readers working on renewable energy, energy-system modelling, climate mitigation and energy policy. Repeated discussions with our colleague at the Sustainable Energy Authority of Ireland (SEAI) have helped shape the study’s policy interpretation and its relevance to Ireland’s offshore hydrogen export ambitions.

An earlier version of this work is available as a preprint at <https://doi.org/10.48550/arXiv.2510.15695>. All authors have reviewed and approved the manuscript, and the authors declare no competing interests. Code and data access details are provided in the manuscript.

This manuscript is not under consideration by any other journal. There are no related manuscripts with overlapping content under consideration or in press elsewhere. We have not had any prior discussions with a Nature Communications editor about this manuscript.

Thank you very much for considering our manuscript.

Yours sincerely,

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